CLASSROOM CONTACT PROGRAMME

(Academic Session : 2025 - 2026)

Enthusiast COURSE

PHASE : ALL PHASE

TARGET : Pre-Medical 2026

Test Type :**MAJOR** Test Pattern :**NEET (UG)**

TEST DATE : 17-05-2026

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	2	1	3	1	4	1	3	2	2	1	3	3	2	3	4	3	3	2	2	2	1	3	3	1	4	1	4	2	4
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	1	3	2	3	3	1	2	1	3	3	4	4	2	3	2	4	2	1	3	3	1	1	1	4	3	3	1	3	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	1	2	4	1	4	4	1	2	4	1	2	1	4	1	2	4	3	4	4	3	1	4	4	3	1	2	1	4	4	4
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	3	3	3	2	1	3	4	3	1	2	3	4	2	2	1	4	2	3	2	3	4	1	3	1	3	1	1	4	4
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	4	4	1	2	1	3	3	2	4	3	2	2	2	1	2	1	2	2	4	2	3	4	4	2	1	2	2	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	2	2	2	2	4	2	1	2	2	4	1	3	3	3	3	4	3	4	3	3	3	1	1	4	1	3	3	1	2

HINT – SHEET

1. Ans (1)

WD = F.d cos 90° = 0

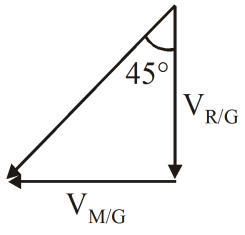
2. Ans (2)

When a car takes a turn on horizontal road, it requires centripetal force to maintain is curved path, so, if it is thrown out in taking a turn, then there is a lack of sufficient centripetal force.

3. Ans (1)

$$\tan 45^\circ = \frac{V_{M/G}}{V_{R/G}} = 1$$

$$V_{R/G} = V_{M/G} = 5 \text{ m/s}$$



4. Ans (3)

Using energy conservation

$$-\frac{GM_Em}{R} + K.E. = 0 + 0$$

$$K.E = \frac{GM_Em}{R} = \frac{GMe}{R^2}.\ mR = mgR.$$

5. Ans (1)

Acceleration due to gravity increases from equator to Poles.

6. Ans (4)

Let the mass of the disc = m

Then translational kinetic energy of the disc

$$K_T = \frac{1}{2}mv^2 \qquad \text{....(i)}$$

When it ascends on a smooth track its rotational kinetic energy will remain same while translational kinetic energy will go on decreasing.

At highest point $K_T = mgh$ or $\frac{1}{2}mv^2 = mgh$

or $h = \frac{v^2}{2g} = \frac{[6]^2}{2 \times 10} = 1.8$ m

7. **Ans (1)**

$$\vec{V}_{com} = \frac{m\vec{v}_1 + m\vec{v}_2}{m+m} = \frac{\vec{v}_1 + \vec{v}_2}{2} = \frac{a\hat{i} + b\hat{j}}{2}$$

$$\vec{a}_{com} = \frac{\vec{a}_1 + \vec{a}_2}{2} = \frac{3}{2}(a\hat{i} + b\hat{j})$$

$\because$ $\vec{v}_{com} \parallel \vec{a}_{com}$, hence path will be straight line.

8. **Ans (3)**

$$\because F = -\frac{dU}{dx} = -\frac{dU}{dx}(a + 2bx^2)$$

$$F = -4bx \Rightarrow K = 4b$$

$$T = 2\pi\sqrt{\frac{m}{4b}} = \pi\sqrt{\frac{m}{b}}$$

9. **Ans (2)**

$(v_1 + v_2) \times 1 = 8$ ..... (1)

$(v_1 - v_2) \times 1 = 0.8$ ..... (2)

by (1) + (2)

$2v_1 = 8.8$

$v_1 = 4.4$ m/s

$v_2 = 3.6$ m/s

10. **Ans (2)**

Horizontal projection from height

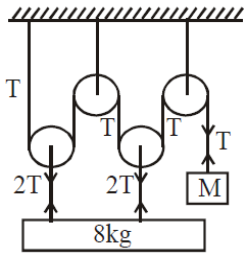
$$R = \sqrt{\frac{2H}{g}}.u \Rightarrow 30 = \sqrt{\frac{2(150-27.5)}{g}}u$$

$$30 = \sqrt{\frac{2 \times 122.5}{9.8}}.u$$

$30 = 5 \times u$

$u = 6$ m/s

11. **Ans (1)**



For 8kg :- $4T = 8g$

$T = 2g$

For M :- $T = Mg$

$2g = Mg$

$M = 2kg$

12. **Ans (3)**

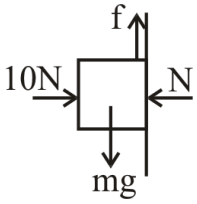
$$[k] = \left[\frac{d^4x}{dt^4}\right] = \frac{[x]}{[T^4]} = \frac{[L]}{[T^4]} = [LT^{-4}]$$

13. **Ans (3)**

Question Explanation: Calculating the weight of a block held stationary against a wall using friction.

Concept: Static Friction

Solution :



N = 10 (In horizontal direction)

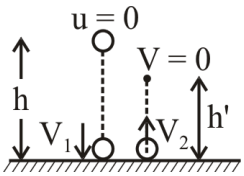
$\Rightarrow$ f = mg (but $f = \mu N$)

$\Rightarrow \mu N = mg \Rightarrow (0.2)(10) = mg$

$\Rightarrow mg = 2N$

Final answer : Option **(3)**

14. **Ans (2)**



Let ball hit floor by V_1 just before first collision.

Energy conservation :

$$\frac{1}{2}mV_1^2 = mgh$$

$$\Rightarrow V_1 = \sqrt{2gh} \quad ....(1)$$

$$e = \frac{V_2}{V_1}$$

[V_2 = Velocity with which ball rises after Ist collision]

Let it attains height h' after first collision.

Energy conservation

$$\frac{1}{2}mV_2^2 = mgh'$$

$$\Rightarrow \frac{1}{2}m(eV_1)^2 = mgh'$$

$$\Rightarrow \frac{1}{2}me^2(2gh) = mgh'$$

$$\Rightarrow h' = e^2h \quad ....(2)$$

So, Total distance travelled before second collision

$= h + 2h'$

$= h + 2e^2h = h(1 + 2e^2)$

15. Ans (3)

$$B = \frac{-P}{\left(\frac{\Delta V}{V}\right)} \Rightarrow \frac{-\Delta V}{V} = \frac{P}{B}$$

$$= \frac{10^5}{1.25 \times 10^{11}} = 8 \times 10^{-7}$$

16. Ans (4)

$$\frac{x - MP}{BP - MP} = \frac{^oC}{100}$$

$$\Rightarrow \frac{x - 39}{239 - 39} = \frac{39}{100}$$

$$\Rightarrow x = 117\ ^\circ W$$

17. Ans (3)

Question Explanation : The problem asks for the effective thermal conductivity (K_{eq}) of a composite cylindrical system consisting of a central solid cylinder (radius R, conductivity K_1) surrounded by a cylindrical shell (outer radius 3R, conductivity K_2). The heat flows lengthwise through both materials simultaneously between two different temperatures.

Concept: Heat Conduction – Parallel combination (composite system)

Formula : Equivalent Conductivity (Parallel):

$$K_{eq} = \frac{K_1A_1 + K_2A_2}{A_{total}}$$

Solution:

The cylinder and shell act as parallel conductors sharing the same temperature difference.

Area of inner cylinder $(A_1) = \pi R^2$

Area of outer shell $(A_2) = \pi(3R)^2 - \pi R^2 = 8\pi R^2$

Total Area $(A) = 9\pi R^2$

Formula: $K_{eq} = \frac{K_1A_1 + K_2A_2}{A}$

$$K_{eq} = \frac{K_1(\pi R^2) + K_2(8\pi R^2)}{9\pi R^2} = \frac{K_1 + 8K_2}{9}.$$

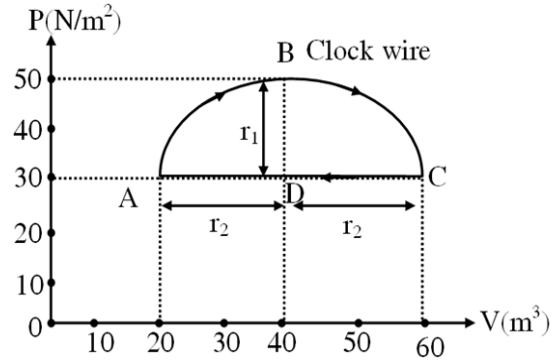
Final Answer: Option(3)

18. Ans (3)

Explain question : Work done by the gas for the cyclic process

ABCDA ⇒

$W_{ABCDA} = ?$



Concept : This question is based on work done in a cyclic process

Formula : Work = Area under the cycle

Solution : Work = Area under the cycle ABCDA

$$= \frac{1}{2} \times \pi r_1 r_2$$

$$= \frac{1}{2} \times \pi \times (50 - 30) \times \left(\frac{60 - 20}{2}\right)$$

$$= \frac{1}{2} \times \pi \times 20 \times \frac{40}{2}$$

$= 200\ \pi$ Joule

∵ The cycle is clock wise in nature

W = +ve

So w = +200 π joule

Final Answer: Option (3)

19. Ans (2)

$$\frac{P}{\rho} = \frac{RT}{M_w}$$

$$\rho = \frac{PM_w}{RT}$$

$$\left[\begin{array}{l} M_w = mN_A \\ \frac{R}{N_A} = k \end{array}\right]$$

$$\rho = \frac{Pm}{kT}$$

20. Ans (2)

$y_1 = a \sin(kx - \omega t)$

$y_2 = a \sin(kx + \omega t)$

According to the principle of superposition, the resultant wave is

$y = y_1 + y_2$

$= a \sin(kx - \omega t) + a \sin(kx + \omega t)$

Using trigonometric identity

$\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$

we get

$y = 2a \sin kx \cos \omega t$

21. Ans (2)

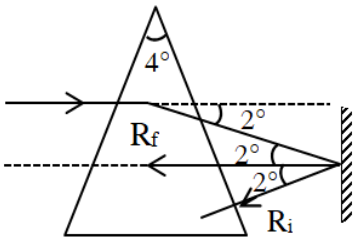
For open organ pipe

$f = \frac{nv}{2\ell} \Rightarrow 1000 = \frac{n \times 330}{2 \times 33} \times 100$

$\Rightarrow 2 = n$

$\therefore f = \frac{2v}{2\ell} \rightarrow 2^{nd}$ harmonic

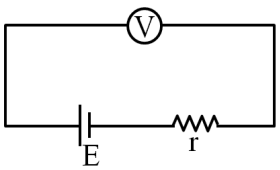
22. Ans (1)



$\delta = (1.5 - 1)\, 4° = 2°$

If mirror is rotated by angle θ then reflected light ray rotated by 2θ. So, if mirror is rotated by 1° reflected light rotate by 2°.

23. Ans (3)



$E = \frac{4 \times 2 + 6 \times 2}{2 + 2} = 5V$

Reading of (V) = 5V

24. Ans (3)

$r = \frac{1}{B}\sqrt{\frac{2mV}{q}}$

$r \propto \sqrt{\frac{m}{q}}$

for H^+, m = 1u , q = +1e

$r_1 \propto 1$

for He^+ , m = 4u , q = +1e

$r_2 \propto \sqrt{\frac{4}{1}} = 2$

for O^{++} , m = 16u , q = +2e

$r_3 \propto \sqrt{\frac{16}{2}}$ \; $= 2\sqrt{2}$

so, $r_1 : r_2 : r_3 = 1 : 2 : 2\sqrt{2}$

25. Ans (1)

Charge particle in uniform magnetic field then time period

$T = \frac{2\pi m}{qB}$

26. Ans (4)

An ideal diode act as a perfect switch

- FB $\rightarrow$ it behaves as short circuit (zero resistance)

  V across diode = 0

  V across R = V

- RB $\rightarrow$ it behaves as open circuit (infinite resistance)

  V across diode = V

  V across R = 0

27. Ans (1)

Shortest wavelength in Balmer series ($\infty \rightarrow 2$)

$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right)(1)^2$

$\frac{1}{\lambda} = \frac{R}{4} \Rightarrow \therefore \lambda = \frac{4}{R} \Rightarrow R = \frac{4}{\lambda}$

Shortest wavelength in Brakett series :-

($\infty \rightarrow 4$)

$\frac{1}{\lambda'} = R\left(\frac{1}{4^2} - \frac{1}{\infty^2}\right)(1)^2$

$\frac{1}{\lambda'} = \frac{R}{16} \Rightarrow \frac{1}{\lambda'} = \frac{4}{\lambda(16)}$

$\therefore \boxed{\lambda' = 4\lambda}$

28. Ans (4)

Frequency of wave not depend on medium while wavelength and velocity depend on medium.

29. Ans (2)

$V_b - V_a = i_1(R_1) - i_2(R_2)$

30. Ans (4)

$$\oint \vec{B}.\vec{d\ell} = \mu_0 I + \mu_0 \epsilon_0 \frac{d\phi E}{dt}$$

31. Ans (1)

$$\lambda_e = \frac{h}{\sqrt{2mE}} \qquad \lambda_p = \frac{hc}{E} \Rightarrow E = \frac{hc}{\lambda_p}$$

$$\therefore \lambda_e = \frac{h\sqrt{\lambda_p}}{\sqrt{2mhc}}$$

$$\therefore \lambda_e = \sqrt{\frac{h\lambda_p}{2mc}}$$

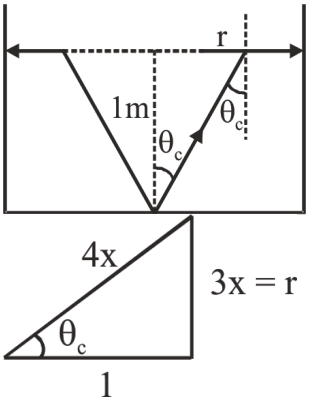
$$\lambda_e \propto \sqrt{\lambda_p}$$

$$\therefore \lambda_e^2 \propto \lambda_p$$

32. Ans (1)

$$\frac{6}{0.125} = R + \frac{1}{\left(\frac{1}{8} + \frac{1}{16} + \frac{1}{16}\right)}$$

33. Ans (3)



$$\sin\theta_c = \frac{3}{4}$$

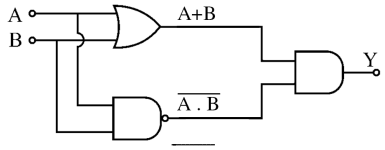
$$\Rightarrow \sqrt{(4x)^2 - (3x)^2} = 1$$

$$\text{or} \quad x = \frac{1}{\sqrt{7}}$$

$$\therefore \quad r = 3x = \frac{3}{\sqrt{7}}$$

$$d = \frac{6}{\sqrt{7}}$$

34. Ans (2)



$Y = (A + B).\overline{(A.B)}$

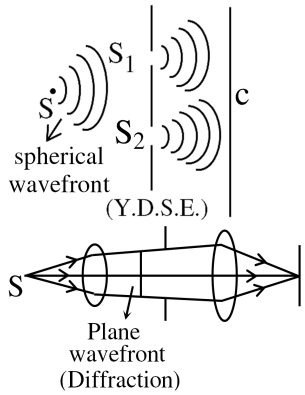
$= (A + B).(\bar{A} + \bar{B})$

$= A\bar{A} + A\bar{B} + B\bar{A} + B\bar{B}$

$= A\bar{B} + B\bar{A}$

= X – OR gate

35. Ans (3)



36. Ans (3)

$$I = \frac{I_0}{2}\cos^2\theta$$

$$\Rightarrow I = \frac{I_0}{2}\cos^2 30°$$

$$= \frac{I_0}{2}\left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{8}I_0$$

37. Ans (1)

- At point A, (inside cavity)

by Gauss law,

$$(4\pi r^2)\,E = \frac{q_{enclosed}}{\varepsilon_0}$$

$$E_A = \frac{kq}{r_A^2} \neq 0$$

- At point B, (inside the metal)

The field within the metal itself is zero. $E_B = 0$

- At point C. (outside the sphere)

Net charge enclosed by gaussian surface through C is +q

$$\therefore E_C = \frac{kq}{r_c^2} \neq 0$$

So, $E_A \neq 0$, $E_B = 0$, $E_C \neq 0$

38. Ans (2)

After connection

$$V_1 = V_2$$

$$\frac{Kq_1^1}{R_1} = \frac{Kq_2^1}{R_2}$$

$$\frac{q_1^1}{q_2^1} = \frac{R_1}{R_2}$$

Here,

q_1^1, q_2^1 - final charge on spheres

$$\frac{\sigma_1}{\sigma_2} = \frac{q_1^1}{4\pi R_1^2} \times \frac{4\pi R_2^2}{q_2^1}$$

$$\frac{\sigma_1}{\sigma_2} = \frac{R_1}{R_2} \times \frac{R_2^2}{R_1^2} = \frac{R_2}{R_1}$$

39. Ans (1)

$$e = \frac{d\phi}{dt} = \frac{dB}{dt}A = A_0\frac{dB}{dt}]$$

$$= A_0\left[\frac{4B_0 - B_0}{t}\right] = \frac{3A_0B_0}{t}$$

40. Ans (3)

$$p_{in} = \frac{880}{88} \times 100$$

$$= 1000 \text{ W}$$

$$I_{input} = \frac{1000}{2200}$$

$$= 0.4545 \text{ A}$$

41. Ans (3)

$$X_L = \omega L = 2000 \times 5 \times 10^{-3} = 10\ \Omega$$

$$X_C = \frac{1}{\omega c} = \frac{1}{2000 \times 50 \times 10^{-6}} = 10\Omega$$

$X_L = X_C$ case of resonance

$$i = \frac{V}{R} = \frac{20/\sqrt{2}}{10} = \sqrt{2}A$$

$$i = 1.4 \text{ A}$$

$$V = iR = 1.4 \times 4 = 5.6 \text{ V}$$

42. Ans (4)

Energy needed to takeout extra neutron from ^{13}C

$Sn = B(^{13}C) - B(^{12}C)$

$13 \times 7.48 - 12 \times 7.68 = 97.24 - 92.16 = 5.08$

$\therefore$ Required energy = 5 MeV

43. Ans (4)

As number of photons incident per second doubles number of electrons emitted also doubles so saturation current doubles.

But $eV_{S1} = h\nu - \phi$

$$eV_{S2} = 2h\nu - 2\phi + \phi$$

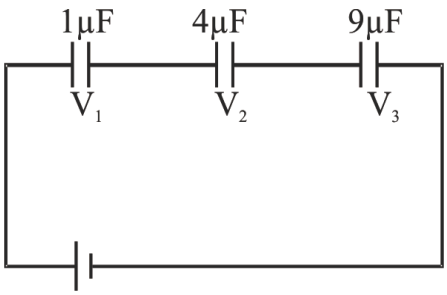
$$eV_{S2} = 2\ eV_{S1} + \phi$$

$$V_{S2} > 2V_{S1}$$

Stopping potential > 20 V

44. Ans (2)

The circuit above can be reduced as drawn in figure below.



In series combination potential is divided in inverse ratio of capacitances so we use the ratio of potential differences across first and second capacitor is

$$\frac{V_2}{V_1} = \frac{1}{4}$$

$$\Rightarrow V_2 = \frac{V_1}{4} = \frac{10}{4} = 2.5V$$

Similarly we use

$$\frac{V_3}{V_1} = \frac{1}{9}$$

$$\Rightarrow V_3 = \frac{V_1}{9} = \frac{10}{9} \text{ V}$$

$$\Rightarrow E = \left(10 + 2.5 + \frac{10}{9}\right) \text{V}$$

45. Ans (3)

In the circuit, C_2 and C_3 are in parallel and their combination is in series with C1.

$C_{||} = C_2 + C_3$

$$C_{eq} = \frac{C_1[C_2 + C_3]}{[C_1 + C_2 + C_3]}$$

If C_3 increases, then C_{eq} increases

$\because$ voltage is fixed

$Q = C_{eq}\, V$

So, Q increases

This Q passes through C_1

So, charge on C_1 increases.

Voltage across | | branch

$$V_{||} = \frac{Q}{C_2 + C_3}$$

$$V_{||} = \frac{C_1 V}{C_1 + C_2 + C_3}$$

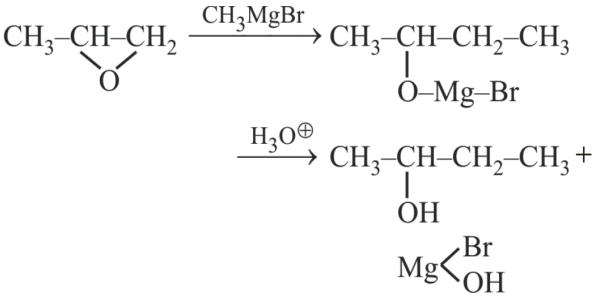
as C_3 increases $V_{||}$ decreases

charge on C_2, $Q_2 = C_2 V_{||}$

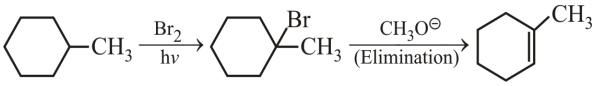
C_2 is fixed and $V_{||}$ decreases,

So, charge on C_2 decreases.

46. Ans (2)



47. Ans (4)



48. Ans (2)

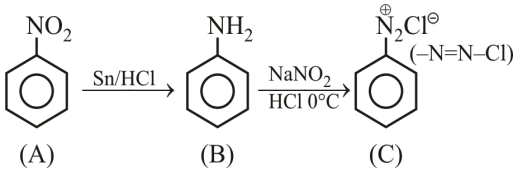
Steric hindrance decrease reactivity of NAR

49. Ans (1)

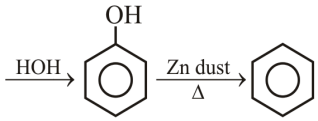
Question Explanation : The question asks to identify the functional groups of the intermediate products A, B, C, and D in the given reaction sequence.

Concept: This question is based on Reduction, Diazotization, Hydrolysis, Reduction (aromatic)

Solution:



(A) (B) (C)



(D)

So, functional groups are: $-NO_2$, $-NH_2$, $-N{=}N-$, $-OH$.

Final Answer: Option (1)

50. Ans (3)

$-NHCOCH_3 \longrightarrow$ +M group

Ortho / Para directing group

51. Ans (3)

chain propagation step

$CH_4 + Cl^{\bullet} \rightarrow \overset{\bullet}{C}H_3 + HCl$

52. Ans (1)

$Fe(OH)_3 \xrightarrow[\text{Excess}]{FeCl_3} Fe(OH)_3|Fe^{+3}, 3Cl^-$

53. Ans (1)

Question Explanation: Degree of dissociation (α) of a weak electrolyte is obtained from its equivalent conductivity at a given concentration and at infinite dilution.

Concept: $\alpha = \Lambda / \Lambda\infty$, % dissociation = $\alpha \times 100$

Solution:

Given:

$\Lambda = 15\ ohm^{-1}\ cm^2\ eq^{-1}$

$\Lambda\infty = 400\ ohm^{-1}\ cm^2\ eq^{-1}$

$\alpha = 15 / 400 = 0.0375$

% dissociation = $0.0375 \times 100 =$ **3.75 %**

Final Answer: Option 1

54. Ans (1)

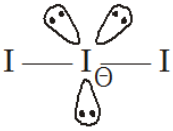


$\Rightarrow \mu_{net} = 0$, but Xe – F bond is polar

55. Ans (4)

M ≡ M ⟶ 0 N ≡ N :⟶ N–

sp, linear sp, linear



sp^3d, Linear



sp^3, Bent

56. Ans (3)

Question Explanation: Identifying the compound that is both paramagnetic and coloured.

Concept: Color, Paramagnetism, and Electronic Configuration.

Solution: The compound must have unpaired d−electrons (d1−9). VOSO4 (V+4,d1) is Paramagnetic and Coloured. **Final Answer**: VOSO4

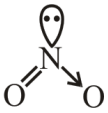
57. Ans (3)

$H_2SO_4 + I^- \longrightarrow I_2$

58. Ans (1)

Chromyl Chloride Test for Cl^- ion

59. Ans (3)



⟶ total $7e^-$ in octate

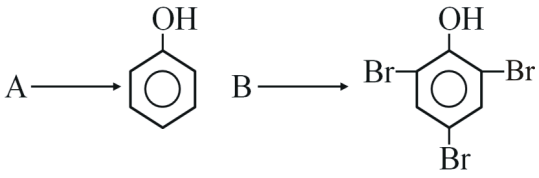
60. Ans (1)

$M(BO_2)_x$ ⟶ where M is transitional metal & x is it's oxidation state.

61. Ans (1)

−ve charge ∝ Back bonding ∝ Strength of M-C bond $\propto \dfrac{1}{\text{Strength of C} - \text{O bond}}$

62. Ans (2)



63. Ans (4)

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64. Ans (1)

Number of orbit = number of wave

65. Ans (4)

Assertion :- Oxidation of iron is given by $4Fe_{(s)} + 3O_2(g) \longrightarrow 2Fe_2O_{3(s)}$ entropy change for rection is −549.4 kJ/mol-K at 298 K. Hence, it is a non-spontaneous process. ⟶ False

Reason :- For a process to be spontaneous, ΔS_{total} & ΔG of a system should be positive & negative respectively. ⟶ True

66. Ans (4)

(i) $NaCl_{(s)} \longrightarrow Na^+_{(aq)} + Cl^-_{(aq)}$ ⟶ $\Delta_{sol}H^0$

(ii) $NaCl_{(s)} \longrightarrow Na^+_{(g)} + Cl^-_{(g)}$ ⟶ Lattice enthalp

(iii) $Na_{(s)} \longrightarrow Na_{(g)}$ ⟶ Enthalpy of atomisation

(iv) $H_{2(g)} + \dfrac{1}{2} O_{2(g)} \longrightarrow H_2O_{(\ell)}$ ⟶ Enthalpy of formation

67. Ans (1)

$Ni(OH)_2 \rightleftharpoons Ni^{+2} + 2OH^-$

s 2s + 0.1

$K_{sp} = (Ni^{+2})\,(OH^-)^2$

$Ni^{+2} = \dfrac{K_{sp}}{(OH^-)^2} = \dfrac{2 \times 10^{-15}}{(0.1)^2} = 2 \times 10^{-13}$

68. Ans (2)

Ideal sol $\Rightarrow \Delta H_{mix} = 0, \Delta V_{mix} = 0$

Non ideal $\Rightarrow \Delta H_{mix} \neq 0, \Delta V_{mix} \neq 0$

Positive deviation $\Rightarrow \Delta H_{mix} > 0, \Delta V_{mix} > 0$

69. Ans (4)

P is called probability factor. $\longrightarrow$ Correct

Z_{AB} represent collision frequency of reactants A & B. $\longrightarrow$ Correct

$e^{-E_a/RT}$ corresponds to the fraction of molecules that have energies equal to or greater than E_a. $\longrightarrow$ Correct

All are correct

70. Ans (1)

$$Kt = 2.303 \log \frac{[A]_0}{[A]_t}$$

$$6 \times t = 2.303 \log \frac{0.5}{0.05}$$

$$t = \frac{2.303}{6} = 0.384 \text{ min}$$

71. Ans (2)

$$\frac{P_A^o - P_S}{P_S} = \frac{\eta_B}{\eta_A}$$

$$\frac{100 - 75}{75} = \frac{w_B \times 18}{60 \times 100}$$

$$\frac{1}{3} = \frac{w_B \times 3}{10 \times 100}$$

$$w_B = \frac{1000}{9} = 111.1\text{g}$$

72. Ans (1)

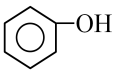
The quantity of solution of Zn^{+2} & Ag^{+} ions

73. Ans (4)

All are correct according to MOT theory.

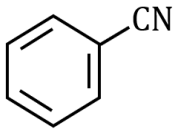
74. Ans (1)

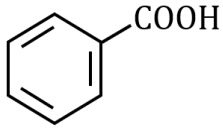
Y is $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$

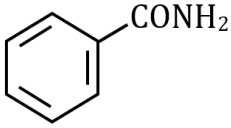
X is 

75. Ans (2)

Final Answer: option (2) $C_6H_5CONH_2$

(A) 

(B) 

(C) 

76. Ans (4)

Brown colour to colourless

77. Ans (3)

(a) The energy of an electron in a multi electron atom depends on its principle quantum number & azimuthal quantum number. $\longrightarrow$ Correct

(b) Energies of the orbitals in the same subshell decreases with increase in the atomic number. $\longrightarrow$ Correct

$E_{2s}(H) > E_{2s}(Li) > E_{2s}(Na) > E_{2s}(K)$ $\longrightarrow$ Correct

(c) Sodium has 10 core electrons $\longrightarrow$ Correct

(d) The size, shape & energy of all 3p orbitals are identical. $\longrightarrow$ Correct

(e) If probability density $|\psi|^2$ is constant on a given surface, $|\psi|$ is also constant over the surface. The boundary surface for $|\psi|^2$ & $|\psi|$ are identical. $\longrightarrow$ Correct

(f) There are two d-orbitals which do not have electron density along the axis. $\longrightarrow$ Incorrect

78. Ans (4)

$NaH_{(s)} + H_2O_{(\ell)} \longrightarrow NaOH_{(aq)} + H_{2(g)}$

79. Ans (4)

$CuSO_4 + 2KCN \rightarrow Cu(CN)_2 \downarrow + K_2SO_4$

$2Cu(CN)_2 \rightarrow 2CuCN \downarrow + (CN)_2 \uparrow$

$CuCN + 3KCN \rightarrow K_3\left[Cu(CN)_4\right]$

80. Ans (3)

Question Explanation: Identify the species that is paramagnetic and shares the calculated bond order of F_2

Concept: This question is based on Molecular orbital theory

Solution: F_2 has 18 electrons and a bond order of 1. B_2 has 10 electrons, a bond order of 1, and two unpaired electrons in $\pi 2p$ orbitals, making it paramagnetic.

$B_2 = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2px^1 = \pi 2py^1$

Final Answer: Option (**3**)

81. Ans (1)

$$\text{Ionic character} \propto \frac{1}{\text{Polarising power of cation}}$$

As positive oxidation state increases or, size of cation decreases, polarizing power of cation increases hence, the correct order of ionic character is :

$MCl > MCl_2 > MCl_3$

82. Ans (4)

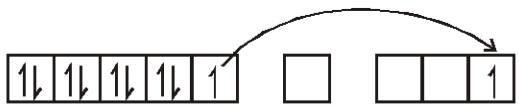
$\mu = NH_3 > NF_3 > BF_3$

$\mu = 1.47\ D\ \mu = 0.24\ D\ \mu = 0$

83. Ans (4)

$[Cu(NH_3)_4]^{+2}$

$Cu^{+2} = 4s^0\ 3d^9$



dsp^2 , paramagnetic

84. Ans (3)

Question Explanation: Find the halide inert to hydrolysis.

Concept: Hydrolysis based on d-orbitals.

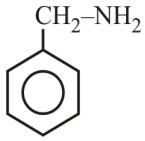
Solution:

$NF_3 + H_2O \longrightarrow$ No reaction

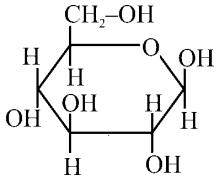
NF3 is inert to hydrolysis at room temperature because nitrogen lacks vacant d-orbitals to accept a lone pair from a water molecule.

Final Answer: (3) NF_3

85. Ans (1)

Duo to localized lone pair  is the strongest base

86. Ans (2)



87. Ans (1)

Assertion : Staggered conformation of ethane is most stable while eclipsed conformation is least stable. $\longrightarrow$ Correct

Reason : Staggered form has the least torsional strain and the eclipsed form has the maximum torsional strain $\longrightarrow$ Correct

Assertion is correct, reason is correct, reason is a correct explanation for assertion

88. Ans (4)

(a) Silica gel can be used as an adsorbent in column chromatography. $\longrightarrow$ Correct

(b) Thin layer chromatography is a type of adsorption chromatography. $\longrightarrow$ Correct

(c) In paper chromatography, the chromatography paper contains water molecules trapped in it. $\longrightarrow$ Correct

89. Ans (4)

$\Delta G_3^0 = \Delta G_2^0 - \Delta G_1^0$

$E_3^0 = 3E_2^0 - -2E_1^0$

$= 0.772$

90. Ans (4)

Question Explanation: We need to derive the expression for rate constant k for the first-order gaseous reaction A ⟶ 3B. where the initial pressure is P_0 and total pressure after time t is P_r.

Concept: This question is based on first-order kinetics and pressure-extent of reaction relationship.

Solution:

Reaction:

A ⟶ 3B

Species	Initial Pressure	Change	Final Pressure
A	P_0	$-P_0\alpha$	$P_0(1-\alpha)$
B	0	$+3P_0\alpha$	$3P_0\alpha$
Total (P_r)	-	-	$P_r = P_0(1+2\alpha)$

From total pressure relation: $\alpha = \frac{P_T/P_0 - 1}{2}$

For a first-order reaction: $kt = \ln\left(\frac{P_0}{P_A}\right)$

Substituting $P_A = P_0(1-\alpha) = \frac{3P_0 - P_r}{2}$:

$k = \frac{1}{t}\ln\left(\frac{P_0}{P_0}\right) = \frac{1}{t}\ln\left(\frac{2P_0}{3P_0 - P_T}\right)$

Final answer: option (4)

91. Ans (2)

Question Explanation:

The question asks to match the events of Prophase I of meiosis with their correct sub-stages.

Concept:

Meiosis I – Prophase I sub-stages: Leptotene, Zygotene, Pachytene, Diplotene, Diakinesis.

Solution: During Prophase I of meiosis:

- Leptotene: Chromosomes begin to condense ⟶ (a ⟶ i)
- Zygotene: Homologous chromosomes pair (synapsis) ⟶ (b ⟶ ii)
- Pachytene: Recombination nodules form and crossing over occurs ⟶ (c ⟶ iv)
- Diplotene: Chiasmata become visible ⟶ (d ⟶ iii)
- Diakinesis: Chromosomes are fully condensed ⟶ (e ⟶ v)

Final Answer: Option (2)

92. Ans (3)

Question Explanation:

The question asks to identify the incorrect statement regarding Anaphase I of meiosis.

Concept:

Meiosis I – Anaphase I (Reductional division)

In this stage, homologous chromosomes separate while sister chromatids remain attached.

Solution:

- (1) Segregation or disjunction of homologous chromosomes occurs ⟶ Correct, homologous chromosomes move to opposite poles.
- (2) Division of centromere is absent ⟶ Correct, centromeres do not split in Anaphase I.
- (3) Chromosome number increase ⟶ Incorrect, because meiosis I is a reductional division, so chromosome number is reduced (not increased).
- (4) Sister chromatids remain associated at their centromere ⟶ Correct.

Final Answer: Option (3)

93. **Ans (3)**

Question Explanation:

The question asks to identify the correct interpretation of population age pyramids and match them with the correct growth patterns.

Concept: Population Ecology – Age pyramids represent the age structure of a population and help determine whether it is expanding, stable, or declining.

Solution:

- Pyramid (i):
  - Broad base (large pre-reproductive population)
  - High birth rate
  - Many young individuals entering reproductive age
  - ⇒ Expanding population
- Pyramid (ii):
  - Narrow base (fewer pre-reproductive individuals)
  - Low replacement of reproductive group
  - Population size decreases over time
  - ⇒ Declining population

Thus:

- (i) ⟶ Expanding
- (ii) ⟶ Declining

This matches row (3) in the given table.

Final Answer: Option (3)

94. **Ans (3)**

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95. **Ans (2)**

Question explanation : Identify the type of symmetry shown in the starfish-like diagram.

Concept: Symmetry

Solution: The correct answer is option 2.

(1) Bilateral symmetry ⟶ Incorrect. Bilateral symmetry means body can be divided into two equal halves only along one
- plane (e.g., humans).

(2) Radial symmetry ⟶ Correct. Starfish and similar echinoderms show radial symmetry, where body can be divided into equal halves along multiple planes passing
- through the central axis.

(3) Spherical symmetry ⟶ Incorrect. Found
- in organisms like Volvox, not starfish.

(4) Asymmetry ⟶ Incorrect. Asymmetry
- means no definite symmetry (e.g., sponges).

The diagram clearly shows radial symmetry.

Final Answer: Option 2

96. **Ans (1)**

Question explanation : Identify the correct matching between common names and scientific names of animals.

Concept : Animal Kingdom

Solution : The correct answer is option (1).

Correct matching is:

A. Silver fish ⟶ Lepisma (ii) Correct.
- Silverfish is an insect (Lepisma).

B. Flying fish ⟶ Exocoetus (iv) Correct.
- Flying fish belongs to genus Exocoetus.

C. Devil fish ⟶ Octopus (i) Correct.
- Octopus is commonly called devil fish.

D. Hag fish ⟶ Myxine (iii) Correct.
- Hagfish belongs to genus Myxine.

Correct matching: A – ii, B – iv, C – i, D – iii.

Final answer : Option (1)

97. **Ans (3)**

Question explanation : Identify the correct definition of tissue.

Concept : Tissue

Solution : The correct answer is option (3) A group of similar cells along with intercellular substances perform a specific function.

A tissue is formed by a group of similar cells and their intercellular material working together to perform a particular function in the body.

- Different types of cells together form organs, not tissue.
- Tissues are specialised for specific functions.

Final answer : Option (3)

98. **Ans (4)**

Question explanation : Identify the labels A, B and C in the diagram of alveoli.

Concept : Respiration

Solution : The correct answer is option 4

- A ⟶ Alveolar cavity : The air-filled space of alveolus where exchange of gases occurs.
- B ⟶ Basement substance : Thin layer present between alveolar wall and capillary wall.
- C ⟶ Red blood cell : RBCs are present inside blood capillaries surrounding alveoli and help in transport of respiratory gases.

Final answer : Option (4)

99. **Ans (3)**

Question explanation : Identify the sources (A and B) from which nitrogenous wastes, water, and ions accumulate in the body.

Concept: Excretion

Solution:

(1) A–Metabolic Activity; B–Excess secretion ⟶ Incorrect. Secretion is not a source of accumulation.

(2) A–Indigestion; B–Excess ingestion ⟶ Incorrect. Indigestion is not relevant here.

(3) A–Metabolic Activity; B–Excess ingestion ⟶ Correct.

- Metabolic activity (A): Nitrogenous wastes like urea, uric acid, ammonia are produced during metabolism of proteins and nucleic acids.
- Excess ingestion (B): Extra salts and water enter the body through diet, leading to accumulation.

(4) A–Metabolic Activity; B–Excess egestion ⟶ Incorrect. Egestion is removal of undigested food, not accumulation.

Correct identification: A = Metabolic Activity, B = Excess ingestion.

Final Answer: Option 3

100. **Ans (1)**

Question explanation : Identify whether the assertion and reason related to myelin sheath and nerve impulse conduction are correct.

Concept : Neural system

Solution : The correct answer is option 1.

- Assertion: Presence of myelin sheath increases the rate of conduction of nerve impulse ⟶ Correct.
- Reason: In myelinated neurons, depolarization occurs only at the nodes of Ranvier, causing saltatory or jumping conduction ⟶ Correct.

Because the impulse jumps from one node to another, conduction becomes much faster compared to non-myelinated neurons. Thus, the reason correctly explains the assertion.

Final answer : Option (1)

101. **Ans (2)**

Question explanation : Identify the feature that differentiates cardiac muscle from skeletal muscle.

Concept : Muscle

Solution : The correct answer is option 2.

Cardiac muscles possess intercalated discs, which are specialized junctions connecting adjacent muscle cells and help in rapid conduction of impulses for coordinated heartbeat.

- Both cardiac and skeletal muscles are striated.
- Sarcolemma and sarcoplasm are present in both types of muscles.

Final answer : Option (2)

102. **Ans (3)**

Question explanation : Identify the hormone that regulates sleep-wake cycle, metabolism and body temperature rhythm.

Concept : Hormones

Solution : The correct answer is option 3.

Melatonin is secreted by the pineal gland and helps regulate circadian rhythms such as sleep-wake cycle, body temperature and daily metabolic activities. Its secretion usually increases during darkness.

- Thyroxine regulates metabolism mainly.
- MSH controls skin pigmentation.
- Insulin regulates blood glucose level.

Final answer : Option (3)

103. **Ans (4)**

Question explanation : Identify the incorrect statement related to hormones.

Concept : Hormones

Solution : The correct answer is option 4

Cortisol is an immunosuppressive hormone and decreases immune response as well as antibody formation. Therefore, statement (4) is incorrect.

- Adrenal cortex hormones like aldosterone stimulate Na^+ reabsorption.
- Thyroid hormones help in regulation of menstrual cycle.
- Estradiol and estrogen act through intracellular/intranuclear receptors.

Final answer : Option (4)

104. Ans (2)

Question explanation : Identify the correct labels for the mammary gland diagram.

Concept: Mammary gland

Solution:

A ⟶ Pectoralis major muscle: The large chest muscle beneath the mammary gland.

B ⟶ Ampulla: Dilated portion of lactiferous duct where milk accumulates.

C ⟶ Areola: Pigmented circular area surrounding the nipple.

D ⟶ Lactiferous duct: Channels that carry milk from lobes to nipple.

Final Answer: Option (2)

105. Ans (2)

Question explanation : Identify the stage during oogenesis when the first polar body is released.

Concept : Oogenesis

Solution : The correct answer is option 2.

During oogenesis, the primary oocyte completes meiosis-I just before ovulation and forms a secondary oocyte along with the first polar body. Thus, the first polar body is released after completion of meiosis-I.

Before birth, primary oocytes remain

- arrested in prophase-I.
- Meiosis-II completes only after fertilisation.

Final answer : Option (2)

106. Ans (1)

Question explanation : Identify the incorrect statements related to immunity and autoimmune diseases.

Concept : Immunity

Solution : The correct answer is option 1

(a) Memory-based immunity evolved in higher vertebrates ⟶ Incorrect, because immunological memory is a characteristic feature of acquired (adaptive) immunity, not innate immunity. Innate immunity is non-specific and does not remember previous infections.

(b) Higher vertebrates can distinguish between self and non-self cells ⟶ Correct. This property helps the immune system protect the body from foreign pathogens.

(c) Rheumatoid arthritis is an autoimmune disease ⟶ Correct, because the immune system attacks the body's own joints.

(d) In autoimmune diseases, body cells attack self cells ⟶ Correct. Due to loss of self-tolerance, immune cells damage normal body tissues.

Therefore, only statement (a) is incorrect.

Final answer : Option (1)

107. **Ans (4)**

Question explanation : Check how many statements about adolescence, parenting, and substance abuse are true.

Concept: Human health & adolescence

Solution:

(A) Alcohol use in adolescence may lead to heavy drinking in adulthood ⟶ True. Early
- exposure increases risk of dependence later.

(B) Drug abuse in adolescence may lead to severe facial and body acne ⟶ True.
Steroid and drug abuse can cause acne due to
- hormonal imbalance.

(C) Parenting with nurturance + discipline lowers risk of substance abuse ⟶ True.
Supportive parenting reduces chances of
- addiction.

(D) Chronic use of drugs/alcohol damages nervous system and liver (cirrhosis) ⟶ True. Long-term abuse causes
- neurodegeneration and liver cirrhosis.

All four statements are true.

Final Answer: Option 4

108. **Ans (2)**

Question explanation : Identify which type of evolution is supported by Stanley Miller's experiment and meteorite evidences.

Concept: Origin of life

Solution: The correct answer is option 2

Stanley Miller performed an experiment showing that simple organic compounds like amino acids could be formed from inorganic
- substances under primitive Earth conditions.

Evidence from meteorites also showed the presence of organic molecules in space. These observations support the theory of chemical evolution, which states that life originated from non-living chemical substances through gradual chemical
- reactions.

Final answer: Option (2)

109. **Ans (3)**

NCERT XI Pg # 115

110. **Ans (2)**

Question explanation : Identify the correct matching among hormones, their source and functions.

Concept : Hormones

Solution : The correct answer is option 2.

Correct matching is:

A. Progesterone ⟶ a. Corpus luteum ⟶ i. Maintain the endometrium

Progesterone maintains uterine endometrium
- for pregnancy.

B. Oestrogen ⟶ c. Theca interna ⟶ iv. Proliferation of endometrium

Oestrogen stimulates growth and
- proliferation of endometrium.

C. Relaxin ⟶ b. Ovary ⟶ ii. Help in parturition

Relaxin relaxes pelvic ligaments and helps
- during childbirth.

D. Oxytocin ⟶ d. Posterior pituitary ⟶ iii. Milk ejection reflex

Oxytocin causes milk let-down reflex and
- uterine contractions.

Final answer : Option (2)

111. **Ans (3)**

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112. **Ans (4)**

Question Explanation: The question is based on the species–area relationship graph and asks which statements are correct regarding slope ((Z)) and species richness ((S)).

Concept: Organisms and Populations – Species Area Relationship

Solution:

Species–area relationship is represented as:

log S = log C + Zlog A Where:

- (S) = Species richness
- (A) = Area
- (Z) = Slope (regression coefficient)
- Greater slope ⟶ steeper line
  Higher position on graph ⟶ greater species
- richness

From the graph:

- Line A is steepest
- Line C is flattest

Therefore: ZA > ZB > ZC

Statement (1) is correct Also, for the same area:

- Line A shows highest species richness
- Line C shows lowest species richness

Thus: SA > SB > SC

Statement (3) is correct

Final Answer: (4) (1) & (3) Both

113. **Ans (1)**

Question Explanation: The question asks to identify the correct statement(s) regarding the endoplasmic reticulum (ER).

Concept: Cell – Endomembrane System

Solution Explanation:

The endoplasmic reticulum (ER) is a membrane-bound organelle involved in synthesis and transport of biomolecules.

- ER is a part of the endomembrane system.
  ER is continuous with the outer nuclear membrane but is not present inside the
- nucleus.
  Rough ER synthesizes proteins, while
- smooth ER synthesizes lipids.

Final Answer: Option (1)

114. **Ans (3)**

Question Explanation: The question asks to identify the incorrect statement regarding the cell cycle.

Concept: Cell Cycle and Cell Division

Solution Explanation:

The cell cycle is a regulated sequence of events involving:

- Cell growth
- DNA replication
- Division of nucleus and cytoplasm

It ensures equal distribution of genetic material to daughter cells.

Final Answer: Option (3)

115. **Ans (1)**

Question explanation : Identify the animal groups showing organ system level, bilateral symmetry and true coelom with body segmentation.

Concept : Animal Kingdom

Solution : The correct answer is option 1

Annelida possess organ system level of organisation, bilateral symmetry, true coelom

- and metameric segmentation.
  Arthropoda are bilaterally symmetrical, coelomate animals with segmented body
- organisation.
  Chordata also show organ system level, bilateral symmetry and true coelom with body
- segmentation.

Mollusca are true coelomates and bilaterally symmetrical but their body is generally unsegmented.

Final answer : Option (1)

116. **Ans (3)**

Question explanation : Identify the junction where hepatic or gastric caeca are present in cockroach.

Concept : Cockroach

Solution : The correct answer is option 3.

In Periplaneta americana, a ring of 6–8 blind tubules called hepatic or gastric caeca is present at the junction of foregut and midgut. These help in digestion and

- secretion of digestive enzymes.

Final answer : Option (3)

117. **Ans (1)**

Question explanation: Identify the correct statement related to lymph.

Concept: Lymphatic Circulation

Solution: The correct answer is option 1

Lymph is a colourless fluid connective tissue derived from blood plasma. It contains the same mineral distribution as plasma because it is formed by filtration of blood through capillary walls.

- Lymph does not contain many RBCs.
- Oxygen concentration in lymph is lower than blood.
- Although lymph is often described as colourless, the option "pale yellow" is not considered the best answer here.

Final answer : Option (1)

118. **Ans (1)**

Question explanation : Identify the major coordinating centre for sensory and motor signalling in brain.

Concept : Neural system

Solution : The correct answer is option 1.

The thalamus acts as the major coordinating and relay centre for sensory and motor signals in the brain. It receives impulses from different sensory organs and transmits them to appropriate regions of cerebrum.

Hypothalamus regulates homeostasis and

- endocrine functions.
- Hippocampus is associated with memory.
- Amygdala is related to emotions.

Final answer : Option (1)

119. **Ans (4)**

Question explanation : Identify the incorrect statement related to PTH (Parathyroid Hormone).

Concept : Hormones

Solution : The correct answer is option 4 .

Parathyroid hormone (PTH) is secreted by parathyroid glands when blood calcium level decreases (hypocalcemia). It increases blood calcium level, therefore it is called a hypercalcemic hormone.

(1) Peptide hormone $\rightarrow$ Correct. PTH is a peptide hormone secreted by parathyroid

- glands.

(2) Hypercalcemic hormone $\rightarrow$ Correct. PTH increases blood calcium levels

- (hypercalcemic effect).

(3) Stimulates bone resorption $\rightarrow$ Correct. PTH activates osteoclasts indirectly,

- increasing Ca^{2+} release from bone.

Hence, statement (4) is incorrect.

Final answer : Option (4)

120. **Ans (4)**

Question explanation : Identify the correct reasons why condoms are popular contraceptives.

Concept : Contraceptive methods

Solution : The correct answer is option 4.

Condoms are widely used contraceptives because:

They act as an effective barrier preventing

- insemination.
- They generally do not interfere with coitus.

They help reduce the risk of sexually transmitted infections (STIs) including

- AIDS.

Thus, all the given statements are correct.

Final answer : Option (4)

121. **Ans (4)**

Question explanation : Identify the disease that can be detected using monoclonal antibodies.

Concept : Cancer

Solution : The correct answer is option 4.

Monoclonal antibodies are highly specific antibodies produced against particular antigens. They are widely used in diagnosis and detection of cancer cells because they can specifically bind to tumour-associated

- antigens.

They are also used in targeted therapy and medical research.

Final answer : Option (4)

122. **Ans (2)**

Question explanation : Check correctness of statements about origin of life and Darwinian theory.

Concept: Evolutionary theories

Solution:

(a) The evolution of life forms has driven by use and disuse of organs ⟶ False. This is Lamarck's theory, not accepted as origin of

- life.

(b) Branching descent and natural selection are the two key concepts of Darwinian Theory ⟶ True. Darwin emphasized common descent with branching and natural

- selection as mechanisms.

Therefore, (a) is false, (b) is true.

Final Answer: Option 2

123. **Ans (4)**

Question explanation : Calculate the number of gametes formed from 100 primary spermatocytes and 100 primary oocytes.

Concept: Gametogenesis

Solution:

- **Spermatogenesis:**

  Each primary spermatocyte undergoes meiosis I ⟶ 2 secondary

  - spermatocytes.

  Each secondary spermatocyte undergoes meiosis II ⟶ 2

  - spermatids.

  Thus, 1 primary spermatocyte ⟶ 4

  - sperms.

  Therefore, 100 primary spermatocytes

  - ⟶ 400 sperms.

- **Oogenesis:**

  Each primary oocyte undergoes meiosis I ⟶ 1 secondary oocyte + 1

  - polar body.

  Meiosis II (completed only at fertilization) ⟶ 1 ovum + 1 polar

  - body.
  - Thus, 1 primary oocyte ⟶ 1 ovum.

  Therefore, 100 primary oocytes ⟶

  - 100 ova.

Correct calculation: 400 sperms & 100 ovum.

Final Answer: Option 4

124. **Ans (4)**

Question Explanation: The question asks to identify the incorrect statement regarding the cytoskeleton.

Concept: Cell – Cytoskeleton **Solution:**

The cytoskeleton is a network of proteinaceous filaments present in the cytoplasm.

Functions:

- Maintains cell shape
- Provides mechanical support
- Helps in cell movement and intracellular transport

Components include:

- Microtubules (made of tubulin)
- Microfilaments
- Intermediate filaments

Final Answer: Option (4)

125. **Ans (1)**

Question Explanation: The question asks to determine the correctness of statements related to meiosis-II.

Concept: Cell Cycle and Cell Division – Meiosis II

Solution:

- In Anaphase-II, sister chromatids separate and move to opposite poles.
- In Prophase-II, chromosomes condense again and nuclear envelope disappears if present.
- Nuclear envelope reappears during Telophase-II, not Prophase-II.

Final Answer: Option (1) Statement-A is correct and B is incorrect

126. **Ans (2)**

Question Explanation: The question asks to identify the incorrect statement regarding the functions of predators in an ecosystem.

Concept: Organisms and Populations – Predation

Explanation:

Predators play important ecological roles:

- Transfer energy across trophic levels
- Control prey population size
- Maintain species diversity by preventing competitive exclusion

Predators generally regulate prey populations rather than completely overexploiting them.

Final Answer: (2) Predators over exploits its prey

127. **Ans (1)**

Question Explanation: The question asks to evaluate the correctness of the assertion and reason related to the pyramid of energy.

Concept: Ecosystem – Energy Flow

Explanation:

Energy flow in an ecosystem follows the laws of thermodynamics.

- At each trophic level, a large amount of energy is lost as heat.
- Therefore, energy available decreases progressively from producers to higher trophic levels.
- Hence, the pyramid of energy is always upright.

Final Answer: (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

128. **Ans (3)**

NCERT XII Pg.No. 263 (15.1.3)

129. **Ans (3)**

Question Explanation: The question asks to determine how many of the given terms are included in the "Evil Quartet" responsible for biodiversity loss.

Concept: Biodiversity and Conservation – Evil Quartet

Explanation: the major causes of biodiversity loss ("Evil Quartet") are:

1. Habitat loss and fragmentation
2. Over-exploitation
3. Alien species invasion
4. Co-extinction

Final Answer: (3) Two

130. **Ans (2)**

Question explanation : Identify the correctly matched animal, phylum and character.

Concept : Animal Kingdom

Solution : The correct answer is option 2.

(1) Jellyfish belongs to Cnidaria but it is commonly found in medusa form, not polyp
- form ⟶ Incorrect.

(2) Fresh water sponge belongs to Porifera
- and has a porous body ⟶ Correct.

(3) Platyhelminthes are acoelomates, not
- pseudocoelomates ⟶ Incorrect.

(4) Mollusca respire mainly through gills
- (ctenidia), not book lungs ⟶ Incorrect.

Therefore, only option (2) is correctly matched.

Final answer : Option (2)

131. **Ans (4)**

Question explanation : Identify which cnidarian exhibits metagenesis (alternation of polyp and medusa forms).

Concept: Cnidaria – life cycle patterns

Solution:

(1) Hydra ⟶ Incorrect. Hydra exists only in
- polyp form, no alternation.

(2) Adamsia (sea anemone) ⟶ Incorrect.
- Sessile polyp only, no medusa stage.

(3) Aurelia (jellyfish) ⟶ Incorrect. Aurelia
- is mainly medusa form, not alternation.

(4) Obelia ⟶ Correct. Obelia shows metagenesis: alternation between polyp
- (asexual) and medusa (sexual) stages.

Thus, Obelia is the correct example of metagenesis among cnidarians.

Final Answer: Option 4

132. **Ans (3)**

Question explanation : Match the labeled sites (A, B, C, D) in the circulatory system diagram with points P and Q on the oxygen–hemoglobin dissociation curve.

Concept: Respiratory physiology – O_2 dissociation curve

Solution:

Point P (high saturation, ~95–100%) ⟶ Represents alveolar blood (lungs, pulmonary
- vein, left atrium/ventricle, systemic arteries).

So A (lungs/alveoli) and B (oxygenated systemic arteries)
- correspond to P.

Point Q (lower saturation, ~70–75%) ⟶ Represents tissues where O_2 is released (systemic veins, right atrium/ventricle,
- deoxygenated blood).

So C (systemic tissues) and D (systemic veins/deoxygenated blood)
- correspond to Q.

Correct match = P = A, B; Q = C, D.

Final Answer: Option 3

133. **Ans (2)**

Question explanation : Identify the correct matching between nephron parts and their physiological functions.

Concept : Excretion

Solution : The correct answer is option 2.

Correct matching is:

A. Proximal convoluted tubule ⟶ iii. Reabsorption of 70–80% of electrolytes

- PCT reabsorbs most of the water, ions and nutrients from filtrate.

B. Distal convoluted tubule ⟶ iv. Ionic balance

- DCT helps in selective secretion and maintenance of ionic and acid-base balance.

C. Henle's loop ⟶ i. Formation of concentrated urine

- Henle's loop establishes osmotic gradient important for concentrating urine.

D. Counter current mechanism ⟶ v. Maintenance of concentration gradient in medulla

- This mechanism maintains high osmolarity in medullary interstitium.

E. Renal corpuscle ⟶ ii. Filtration of blood

- Glomerulus and Bowman's capsule together filter blood plasma.

Final answer : Option (2)

134. **Ans (2)**

Question explanation : Identify the number of correct statements related to human RBCs.

Concept : Circulation

Solution : The correct answer is option 2.

(A) Incorrect ⟶ Normal RBC count in human females is about 4.5–5 million/mm³,

- not 1.5–3.0 million/mm³.

(B) Correct ⟶ RBCs are the most abundant

- cells in blood.

(C) Incorrect ⟶ Human RBCs are biconcave and non-nucleated, not oval. Oval

- RBCs are found in camel.

(D) Correct ⟶ Average life span of RBCs is

- about 120 days.

Therefore, only statements B and D are correct.

Final answer : Option (2)

135. **Ans (2)**

Question explanation : Identify the correct statements related to muscle contraction.

Concept : Muscle contraction

Solution : The correct answer is option 2.

A. I bands get reduced ⟶ Correct, because actin filaments slide inward during

- contraction.

B. A band retain the length ⟶ Correct, as the length of myosin filaments does not

- change.

C. Length of sarcomere decreases ⟶

- Correct, due to sliding filament mechanism.

D. Breakdown of ATP ⟶ Correct, ATP

- provides energy for muscle contraction.

E. H-zone remain unchanged ⟶ Incorrect, because H-zone decreases or may disappear

- during contraction.

Thus, statements A, B, C and D are correct.

Final answer : Option (2)

136. **Ans (1)**

Question explanation : Identify whether the statements related to drug and alcohol abuse are correct.

Concept : Drug and alcohol abuse

Solution : The correct answer is option 1.

Statement-I: Drug and alcohol abuse can have far-reaching implications ⟶ Correct. It affects physical health, mental health, family life and society.

Statement-II: The adverse effects are not limited only to the user ⟶ Correct. Substance abuse can also affect family members, unborn children and social relationships.

Thus, both statements are correct.

Final answer : Option (1)

137. **Ans (2)**

Question Explanation: The question asks to identify the statement that is not true regarding trophic levels and energy flow in ecosystems.

Concept: Ecosystem – Trophic Levels

Explanation:

Organisms may occupy more than one trophic level depending on food habits.

Primary consumers are herbivores feeding on producers.

Energy decreases at successive trophic levels.

Detritivores feed on dead organic matter originating from all trophic levels, including producers.

Final Answer: Option (2)

138. **Ans (1)**

Question Explanation: The question asks to evaluate the correctness of the assertion and reason related to population ecology and evolution.

Concept: Organisms and Populations

Solution:

Population ecology studies populations in relation to the environment and connects ecology with genetics and evolution.

Natural selection acts on populations by favoring individuals with advantageous traits, leading to evolutionary changes over generations.

Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

Final Answer: Option (1)

139. **Ans (2)**

Question Explanation:

The question asks to determine the percentage of pure plants in the F_2 generation of Mendel's monohybrid cross.

Concept: Principles of Inheritance – Monohybrid Cross

Explanation:

In a monohybrid cross: Tt * Tt

F_2 genotypic ratio: 1 TT : 2 Tt : 1 tt

Pure plants are homozygous: (TT), (tt)

Thus, pure plants = (1 + 1 = 2) out of 4

Percentage of pure plants: $\frac{2}{4} * 100 = 50\%$

Final Answer: (2) 50%

140. **Ans (2)**

Question Explanation: The question asks to evaluate the correctness of the assertion and reason related to inheritance of haemophilia.

Concept: Principles of Inheritance – Sex-linked Inheritance

Explanation:

Haemophilia is an X-linked recessive disorder.

- Fathers pass their Y chromosome to sons.
- Sons receive the X chromosome from their mother.
- Therefore, a father cannot pass the haemophilia gene directly to his sons.

Final Answer: (2) Both (A) and (R) are correct and (R) is the correct explanation of (A)

141. **Ans (4)**

Question Explanation: The question asks to evaluate the correctness of the assertion and reason related to the universality of the genetic code.

Concept: Molecular Basis of Inheritance – Genetic Code

Explanation:

The genetic code is called nearly universal because the same codon generally specifies the same amino acid in almost all organisms.

Exceptions occur in:

- Mitochondria
- Some protozoans

where a few codons may code differently.

The reason correctly explains why the code is termed "nearly" universal rather than absolutely universal.

Final Answer: (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

142. **Ans (2)**

Question Explanation: The question asks to identify the correct statement related to microbes and their industrial uses.

Concept: Microbes in Human Welfare

Explanation:

Different microorganisms are used in industries and medicine:

- Aspergillus niger is a fungus used for citric acid production.
- Streptokinase is used as a clot buster.
- Large holes in Swiss cheese are produced by Propionibacterium.
- Toddy is produced by fermentation using yeast.

Final Answer: (2) Streptokinase is used as a clot buster

143. **Ans (3)**

Question Explanation: The question asks to identify the incorrect statement about satellite DNA (VNTRs).

Concept: – DNA Fingerprinting **Solution:**

VNTRs (Variable Number Tandem Repeats) are:

- Repetitive, non-coding DNA sequences
- Present in tandem repeats
- Highly polymorphic
- Used in DNA fingerprinting
- Do not code for proteins

Final Answer: Option (3)

144. **Ans (4)**

Question Explanation: The question asks to identify the allele pair that will produce a recessive phenotype based on enzyme functionality in converting substrate S into product P.

Concept: – Gene Expression & Enzyme Function

Explanation:

A dominant phenotype is produced when at least one allele produces a functional or partially functional enzyme that converts substrate S $\longrightarrow$ P.

A recessive phenotype appears only when no functional enzyme is produced.

- A, a, $\alpha \longrightarrow$ produce functional or partially functional enzymes $\longrightarrow$ dominant phenotype
- $\beta \longrightarrow$ non-functional enzyme
- $\gamma \longrightarrow$ no enzyme produced

Final Answer: (4) β, γ

145. **Ans (4)**

Question Explanation: The question asks which enzyme(s) are used for clarification of bottled fruit juices.

Concept: Microbes in Human Welfare – Industrial Enzymes

Explanation:

Fruit juice clarification involves removal of haze-causing substances like pectin, proteins, and other suspended particles.

- Pectinases break down pectin ⟶ improves clarity
- Proteases break down proteins ⟶ also helps in removing haze
- Streptokinase is a clot-buster enzyme, not used in food processing

Final Answer: (4) Both (1) and (2)

146. **Ans (2)**

Question Explanation: The question asks to identify the correct statements about RNA interference (RNAi).

Concept: Molecular Basis of Inheritance – RNA Interference

Solution:

RNA interference (RNAi) is a natural gene-silencing mechanism in eukaryotes where specific mRNA is inhibited or degraded by complementary RNA molecules. It also plays a role in defense against viruses and is used in biotechnology for pest resistance in plants.

Final Answer: Option (2)

147. **Ans (1)**

Question Explanation: The question asks to evaluate the correctness of the Assertion and Reason related to the transgenic cow "Rosie".

Concept: Biotechnology – Transgenic Animals

Solution:

Transgenic animals are genetically modified to express foreign genes that improve desirable traits. Rosie (transgenic cow) was developed to produce milk containing human protein for better infant nutrition.

Since human α-lactalbumin is the key reason for improved nutritional quality, the reason correctly explains the assertion.

Final Answer: Option (1)

148. **Ans (2)**

Question Explanation: The question asks which government body regulates GM (genetically modified) research and ensures safety of GM organisms before public use in India.

Concept: Biotechnology – Regulation of GMOs in India

Solution:

In India, the approval and safety assessment of genetically modified organisms (GMOs) is handled by regulatory bodies. The key authority responsible for approval of GM research and release is the Genetic Engineering Approval Committee (GEAC).

Final Answer: Option (2)

149. **Ans (2)**

Question Explanation: The question asks to identify the correct labeling of a transcription unit based on its schematic structure.

Concept: Molecular Basis of Inheritance – Transcription Unit

Solution: A transcription unit consists of:

- Promoter ⟶ site for RNA polymerase binding (start of transcription)
- Structural gene ⟶ region transcribed into RNA
- Terminator ⟶ signals end of transcription

Strands:

- Coding strand (sense strand) ⟶ same sequence as RNA (5' ⟶ 3')
- Template strand (antisense strand) ⟶ used for RNA synthesis (3' ⟶ 5')

(A) Coding strand

(B) Promoter

Thus: (C) Terminator **Final Answer:** Option (2)

150. **Ans (2)**

Question Explanation: The question asks what property DNA acquired after evolving from RNA due to chemical modifications.

Concept: Molecular Basis of Inheritance **Solution:** RNA is considered the first genetic material. DNA evolved later from RNA with chemical modifications (like deoxyribose sugar and thymine instead of uracil), which made it more chemically stable.

DNA is therefore better suited for long-term storage of genetic information.

Final Answer: Option (2)

151. **Ans (3)**

Question Explanation: The question asks the number of types of gametes produced by a diploid organism heterozygous at 4 loci.

Concept: Principles of Inheritance – Gamete Formation

Solution:

Number of types of gametes formed is given by: 2n where (n) = number of heterozygous loci.

Here, (n = 4)

So, $2^4 = 16$

Possible gametes = 16 types

Final Answer: Option (3)

152. **Ans (2)**

Question Explanation: The question asks to identify the term used for the small opening in the DNA helix where replication occurs.

Concept: Molecular Basis of Inheritance – DNA Replication

Solution:

During DNA replication, the double helix unwinds at a specific point forming a Y-shaped structure where new strands are synthesized.

This structure is called the replication fork.

Final Answer: Option (2)

153. **Ans (2)**

Question Explanation: The question asks why recombinant bacterial colonies appear white in blue-white screening during gene cloning.

Concept: Biotechnology – Recombinant DNA Technology

Solution: In blue-white screening:

- The lacZ gene encodes β-galactosidase.
- This enzyme cleaves X-gal to produce a blue color.
- Insertion of foreign DNA disrupts (inactivates) the lacZ gene ⟶ no enzyme formed ⟶ colonies remain white.

Final Answer: Option (2)

154. Ans (2)

Question Explanation: The question asks which pest is controlled by the protein product of the Bt toxin gene cry IAb.

Concept: Biotechnology – Bt Crops

Solution:

Bt toxin genes (from Bacillus thuringiensis) produce insecticidal proteins (Cry proteins) that are specific to particular insect pests.

cry IAb is mainly effective against corn
- borer (Ostrinia species) in maize.

Final Answer: Option (2)

155. Ans (2)

Question Explanation: The question asks to identify the incorrect statement regarding Mendel's monohybrid cross.

Concept: Principles of Inheritance – Monohybrid Cross

Solution: In Mendel's monohybrid cross:

F_1 generation shows only dominant trait (no
- blending)
- F_2 generation shows 3:1 phenotypic ratio
- Traits do not blend in F_1 or F_2
- Tall (T) is dominant over dwarf (t)

Final Answer: Option (2)

156. Ans (4)

Question Explanation: The question asks to fill in the blanks about the evolutionary relationship between RNA and DNA and their stability.

Concept: Molecular Basis of Inheritance – Origin of Genetic Material

Solution:

RNA is considered the first genetic material in evolution. DNA evolved later from RNA through chemical modifications such as:

- Replacement of ribose with deoxyribose
- Presence of thymine instead of uracil

These changes make DNA more chemically stable than RNA.

Final Answer: Option (4)

157. Ans (2)

NCERT XII Pg. # 197

158. Ans (1)

NCERT XII Pg.# 111

159. Ans (2)

Question Explanation: The question asks about recombination frequency in Morgan's linkage experiments in Drosophila.

Concept: Linkage and Recombination – Morgan's Experiments

Solution:

In linked genes:

Tightly linked genes ⟶ very low
- recombination frequency

Loosely linked genes ⟶ higher
- recombination frequency

Recombination frequency indicates distance
- between genes on a chromosome

Cross I (tightly linked genes)
- Genes are very close on X-chromosome
- Very low crossing over
- Recombinants = 1.3%

Cross II (loosely linked genes)
- Genes are farther apart
- Higher crossing over
- Recombinants = 37.2%

Final Answer: Option (2)

160. Ans (2)

Question Explanation: The question asks the probability of a carrier child when both parents are carriers of sickle cell anaemia.

Concept: Sickle cell anaemia

Solution:

Sickle cell anaemia shows autosomal recessive inheritance.

Let:
- HbA = normal allele
- HbS = sickle cell allele

Probability of carrier child: 24*100 = 50%

Final Answer: Option (2)

161. Ans (4)

NCERT XIth Pg.#17 Table-2.1

162. Ans (1)

NCERT-XI Pg # 32, 35, 34

163. Ans (3)

NCERT XI Pg # 38/39

164. Ans (3)

Question Explanation: The question asks to identify the correct statement related to the electron transport chain (ETC) in mitochondria.

Concept: Cellular Respiration – Oxidative Phosphorylation

Solution: In mitochondria:

- Cytochrome c is a small protein located in the intermembrane space, loosely associated with the inner mitochondrial membrane.
- It transfers electrons between Complex III and Complex IV.
- Complex IV is called cytochrome c oxidase.
- Complex III contains cytochromes b and c_1 along with iron-sulfur centers.

Final Answer: Option (3)

165. Ans (3)

Question Explanation: The question asks to identify the correct match related to fertilization events in angiosperms.

Concept: Sexual Reproduction in Flowering Plants – Double Fertilization

Solution:

In angiosperms, two fusion events occur:

- Syngamy: male gamete + egg cell ⟶ zygote
- Triple fusion: male gamete + secondary nucleus (polar nuclei fusion product) ⟶ primary endosperm nucleus (endosperm formation)

Final Answer: Option (3) gamete + Secondary nucleus ⟶ Triple fusion

166. Ans (3)

NCERT-XI, Pg. # 154 to 158

167. Ans (4)

Question Explanation: The question asks to identify the incorrect statement about conditions required for plant growth.

Concept: Plant Growth and Development **Solution:** Growth in plants depends on several environmental and physiological factors:

- Oxygen is required for respiration to release energy
- Light and gravity act as environmental signals affecting growth
- Every organism has an optimum temperature range for growth
- Enzymatic reactions require water as a medium

Final Answer: Option (4)

168. Ans (3)

Question Explanation: The question asks about the chemical composition of the cell wall in chrysophytes.

Concept: Biological Classification – Protista

Solution:

Chrysophytes (golden algae, diatoms) are mainly aquatic microorganisms. Their cell wall (especially in diatoms) is known as a frustule, made of silica.

Final Answer: Option (3)

169. Ans (4)

Question Explanation: The question asks to identify the correct labeling of different regions of a root tip based on its structure.

Concept: Morphology of Flowering Plants – Root Zones

Solution:

A typical root tip shows four main regions in order from apex upward:

- Root cap ⟶ protective structure at the tip
- Meristematic region ⟶ active cell division
- Elongation region ⟶ cells elongate and increase root length
- Maturation region ⟶ cells differentiate; root hairs are present

Final Answer: Option (4)

170. Ans (3)

NCERT XI, Pg.# 145

171. Ans (3)

NCERT XIth, Page : 22, Para-2.3

172. Ans (3)

NCERT Pg # 39

173. Ans (1)

Question Explanation: The question asks to identify the correct labelling of fruit parts in mango and coconut (drupe type fruits).

Concept: Morphology of Flowering Plants – Fruit Structure

Solution:

Both mango and coconut are drupes, which have a pericarp differentiated into:

- Epicarp ⟶ outer skin
- Mesocarp ⟶ fleshy in mango, fibrous in coconut
- Endocarp ⟶ hard stony layer around seed
- Seed ⟶ inner reproductive unit

Final Answer: Option (1)

174. Ans (1)

Question Explanation: The question asks to evaluate the correctness of the Assertion and Reason related to the phases of photosynthesis.

Concept: Photosynthesis – Light and Dark Reactions

Solution: Photosynthesis has two phases:

- Light reaction ⟶ produces ATP and NADPH
- Biosynthetic phase (Calvin cycle / dark reaction) ⟶ uses ATP and NADPH to fix CO_2

The biosynthetic phase does not directly require light but depends on ATP and NADPH produced in the light reaction.

Final Answer: Option (1)

175. Ans (4)

Question Explanation: The question asks to identify the incorrect statement(s) related to photosynthesis and chloroplast behavior.

Concept: Photosynthesis – Light Reaction & Pigments

Solution:

- Chloroplasts can change orientation depending on light intensity (photorelocation).
- Photosynthesis involves multiple pigments, not just chlorophyll a.
- Action spectrum and absorption spectrum are similar but not identical.

Final Answer: Option (4)

176. Ans (1)

Question Explanation: The question asks to complete a table based on plant classification features of bryophytes, pteridophytes, and gymnosperms.

Concept: Plant Kingdom – Alternation of Generations & Reproduction

Solution:

- Bryophyta: dominant gametophyte, no vascular tissue, female sex organ = archegonium
- Pteridophyta: dominant sporophyte, vascular tissue present, female sex organ = archegonium
- Gymnosperms: dominant sporophyte, vascular tissue present, female sex organ = archegonium

Fertilization:

- Bryophytes & pteridophytes ⟶ zoidogamy (motile sperm)
- Gymnosperms ⟶ siphonogamy (pollen tube)

Final Answer: Option (1)

177. **Ans (3)**

NCERT Page : 90, 91

178. **Ans (3)**

Question Explanation: The question asks to identify the incorrect taxonomic statement.

Concept: Taxonomic hierarchy

Solution:

Taxa are arranged in hierarchy: species ⟶ genus ⟶ family ⟶ order ⟶ class ⟶ phylum ⟶ kingdom.

Key facts:

- Carnivora is an order, not a family
- Diptera is an order (flies), not a family
- Mango belongs to family Anacardiaceae

Final Answer: Option (3)

179. **Ans (1)**

Question Explanation: The question asks the effect of spraying gibberellin (GA) on juvenile conifers.

Concept: Plant Growth Regulators – Gibberellins

Solution: Gibberellins are plant hormones that:

- Promote stem elongation
- Break dormancy
- Induce flowering in certain plants
- Help overcome juvenile phase in some gymnosperms like conifers

In juvenile conifers, application of gibberellins helps in:

- Reducing the long juvenile phase
- Inducing early reproductive maturity (early seed formation)

So, GA treatment leads to earlier seed bearing in conifers.

Final Answer: Option (1)

180. **Ans (2)**

Question Explanation:

The question asks about the step in the Krebs cycle in which succinyl-CoA is formed. **Concept:** Krebs Cycle (Citric Acid Cycle) – Oxidative decarboxylation reactions during aerobic respiration.

Solution:

In the Krebs cycle, succinyl-CoA is formed from α-ketoglutarate in a key oxidative decarboxylation step. During this reaction:

- One molecule of CO_2 is released
- NAD^+ is reduced to NADH + H^+
- Coenzyme A joins to form succinyl-CoA (4-carbon compound)

This occurs after the first decarboxylation step (isocitrate ⟶ α-ketoglutarate). Thus, succinyl-CoA formation is associated with the second oxidative decarboxylation step of the cycle. The phrase "two successive decarboxylations" refers to the overall sequence starting from citric acid ⟶ isocitrate ⟶ α-ketoglutarate ⟶ succinyl-CoA.

Final Answer: Option (2)